Chapter 7: Behavioral Strategies

ECON 5001
Prof. Paul J. Healy
Based on: *Game Theory* by Giacomo Bonanno

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Where We Left Off

Chapter 4: game trees with **information sets**, solved by SPNE

- Solve the smallest subgames first, work backwards
  - A generalization of backwards induction
- We said if a subgame has **no** NE, then there's **no SPNE**

Chapter 6: **mixed strategies**

- **Nash's Theorem**: every finite game (or subgame) has a (possibly mixed) NE
- So the "no NE in a subgame" problem can be fixed up
- But we have to allow mixing inside a game tree

Chapter 7:

1. Put lotteries and mixing into game trees
2. Find a **simpler** way to mix in trees: behavioral strategies
3. Show that a SPNE always exists
4. See what SPNE still gets wrong

Lotteries in Game Trees

“The Game” 2013

2013 OSU @ Michigan. Urban Meyer vs. Brady Hoke. 1,129 total offensive yards
Was 42-35. With 32 sec left, MI scores TD. Now 42-41.

- Brady Hoke: **Kick** to tie/OT or **Go for 2** to win

- **Kick**: Good 95% of the time, almost surely leads to overtime
  - Overtime: let’s say it’s 50-50 who wins
- **Go for 2**: Model it as a 2×2 subgame
  - Michigan: **Run** or **Pass**
  - OSU: **Stop Run (SR)** or **Stop Pass (SP)**

Outcome Payoffs: $u(\text{Win}) = 100$, $u(\text{Lose}) = 0$ for both teams

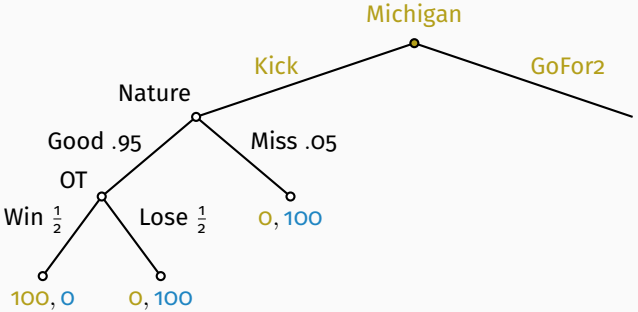
How to draw this tree?

Drawing the Tree



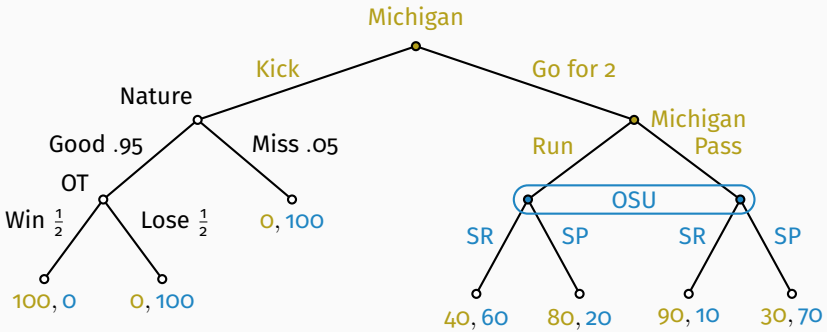
Michigan moves first

Drawing the Tree



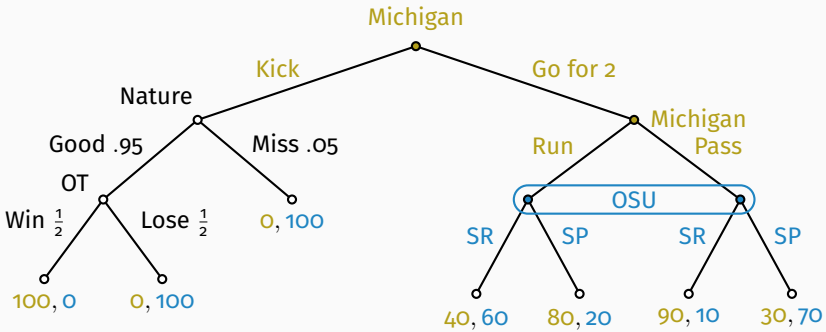
Kick: two moves by “Nature”. Each branch has a probability

Drawing the Tree



GoFor2: a simultaneous-move game hanging off the tree

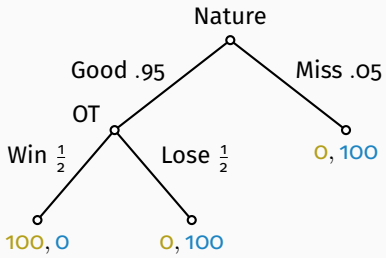
Drawing the Tree



How many proper subgames? (Include moves by Nature.) _____

Nature Gives a Lottery Over Outcomes

After Michigan chooses Kick:



Overtime gives Michigan $\begin{pmatrix} \text{Win} & \text{Lose} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$U(\text{OT}) =$ _____

Kick gives Michigan $\begin{pmatrix} \text{OT} & \text{Lose} \\ .95 & .05 \end{pmatrix}$

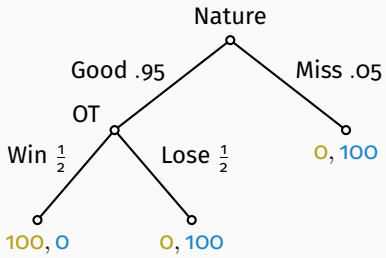
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Rule: Same as Chapter 6. Use expected utilities

Bonanno (§7.1) writes the lottery as the payoff instead of drawing Nature. Using Nature is more standard.

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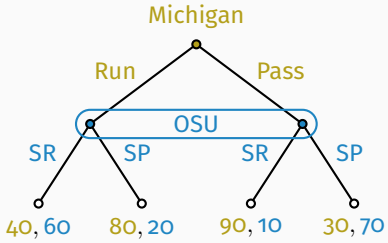
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So: replace the entire Kick branch with _____

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Subgame-Perfect Equilibrium: Allowing Mixing

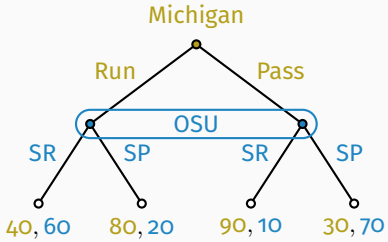
Step 1: Solve the Subgame After “Go for 2”



Michigan \ OSU	Stop Run	Stop Pass
Run	40, 60	80, 20
Pass	90, 10	30, 70

Payoffs = Expected utility = Prob. of Win

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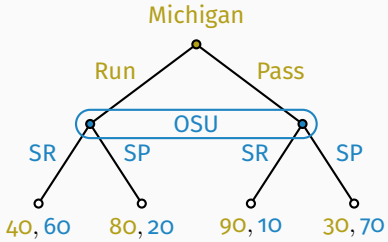


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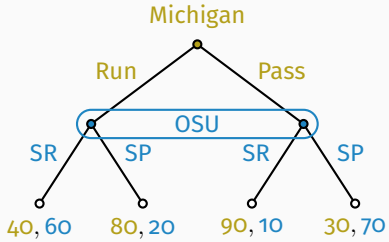


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Chapter 7: let's include subgame MSNE in our definition of SPNE.

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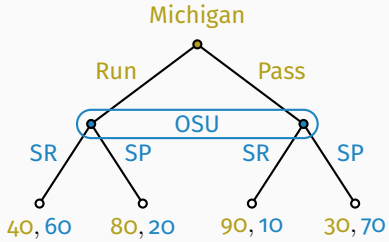


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Michigan's p on Run that makes OSU indifferent:
$U(SR) = 60p + 10(1-p)$ $U(SP) = 20p + 70(1-p)$ $U(SR) = U(SP) \Rightarrow p^* = 3/5$

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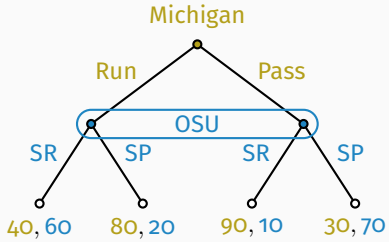


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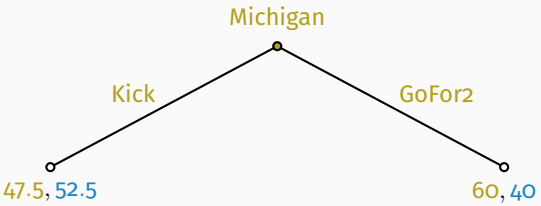
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Expected Payoffs in MSNE:
Michigan: 60 (percent chance of winning)
OSU: 40 (percent chance of winning)

Step 2: Replace and Solve



Kick: replace that subgame by its expected utility
Go for 2: the subgame, replaced by its **mixed** NE payoffs

Michigan chooses: _____

What Actually Happened in 2013



Photo: Shelby Lum, *The Lantern*

The Decision:
Hoke went for 2.
Consistent with SPNE!

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The Play:
Michigan passed
Tyvis Powell (23) intercepted
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Did Hoke make a bad choice?

What Actually Happened in 2013



Photo: Shelby Lum, *The Lantern*

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Hoke went for 2.
Consistent with SPNE!

The Play:
Michigan passed
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OSU wins 42–41

Did Hoke make a bad choice?
A 60% chance still
loses 40% of the time

The SPNE

The SPNE is fully specified by giving the probability of every action at **every** decision node:

$$\text{Michigan: } \sigma_1^* = \left(\begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ \underline{\quad} & \underline{\quad} & \underline{\quad} & \underline{\quad} \end{array} \right) \qquad \text{OSU: } \sigma_2^* = \left(\begin{array}{cc} \text{SR} & \text{SP} \\ \underline{\quad} & \underline{\quad} \end{array} \right)$$

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Notice: we didn't write a probability for each **strategy**
We wrote one probability distribution per **information set**

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That's called a **behavioral strategy**

Selten's Theorem

Theorem (Selten, 1965)
Every finite extensive-form game with cardinal payoffs has **at least one** SPNE (possibly involving mixed strategies).

(**Finite game:** Finite # players, finite # of actions, tree doesn't go forever.)

Why?

- Every subgame is a finite game
- Nash: every finite game has a (mixed) NE
- So the SPNE algorithm can give a prediction at every subgame
- And the game is finite, so the algorithm doesn't run forever

Reinhard Selten shared the 1994 Nobel Prize with John Nash and John Harsanyi.

Behavioral Strategies

Michigan's Strategies

A strategy: one choice at every information set (Chapter 4)

Michigan has two information sets, two choices each:

$$S_{MI} = \{\ (\text{Kick}, \text{Run}),\ (\text{Kick}, \text{Pass}),\ (\text{GoFor2}, \text{Run}),\ (\text{GoFor2}, \text{Pass})\ \}$$

- (Kick, Run) and (Kick, Pass) are different strategies, even though Kick precludes the Run/Pass decision.

Mixed strategy (Chapter 6): a probability for each of the 4 strategies

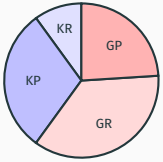
$$\begin{pmatrix} (\text{Kick}, \text{Run}) & (\text{Kick}, \text{Pass}) & (\text{GoFor2}, \text{Run}) & (\text{GoFor2}, \text{Pass}) \\ p & q & r & 1-p-q-r \end{pmatrix}$$

- Needs _____ numbers

Two Ways To Randomize

Mixed strategy

- Before the game: spin one big wheel over **complete plans**
- Then follow the plan



Behavioral strategy

- At each information set, when you get there: flip that set's coin
- One coin per information set



Behavioral strategy: $\left(\begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ p & 1-p & q & 1-q \end{array}\right)$ Needs _____ numbers

Formal Definition: Behavioral Strategy

Definition (Bonanno 7.1.1)
A **behavioral strategy** for a player is a list of probability distributions, one for every information set of that player. These give the probability the player plays each action at each information set.

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Fact: If a player has only one information set then mixed $=$ behavioral (like OSU)

Are They the Same?

Suppose Michigan's coach uses this mixed strategy:

$$\sigma_1 = \begin{pmatrix} (\text{Kick}, \text{Run}) & (\text{Kick}, \text{Pass}) & (\text{GoFor2}, \text{Run}) & (\text{GoFor2}, \text{Pass}) \\ 0.1 & 0.3 & 0.36 & 0.24 \end{pmatrix}$$

Find a behavioral strategy that does the same thing:

$\Pr(\text{Kick}) =$ _____ $\Pr(\text{GoFor2}) =$ _____

$\Pr(\text{Run} \mid \text{GoFor2}) =$ _____ $\Pr(\text{Pass} \mid \text{GoFor2}) =$ _____

Notation: $\Pr(\text{A} \mid \text{B})$ means "Probability of A, given *B* already occurred."
This is called a *conditional probability*.

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Check: $\Pr(\text{GoFor2 and Run}) =$ _____ $= 0.36$ ✓

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Check: $\Pr(\text{GoFor2 and Run}) =$ _____ $= 0.36$ ✓

Lesson: The 0.1 vs. 0.3 split never matters. The behavioral strategy doesn't require it.

Kuhn's Theorem

Theorem (Kuhn, 1953)
In games with **perfect recall**, for every mixed strategy there is a behavioral strategy that gives the same probability over terminal nodes

Recall perfect recall (Chapter 4):

- Never forget what you knew
- Never forget what you did

Takeaway: Use behavioral strategies. They're simpler, and nothing is lost

Notation: We'll use σ_i to describe either a mixed or behavioral strategy

Bonanno works through his own example (Figure 7.3) on pp. 239–241.

Two Ways To Be Unpredictable

Randomize **as you go** (behavioral):

- **Play-calling**: each down, run or pass?
- **Pitching**: each pitch, fastball or curveball?
- **Poker**: each betting round, bluff or fold?
- **Penalty kicks**: each kick, left or right?

Randomize **a whole plan in advance** (mixed):

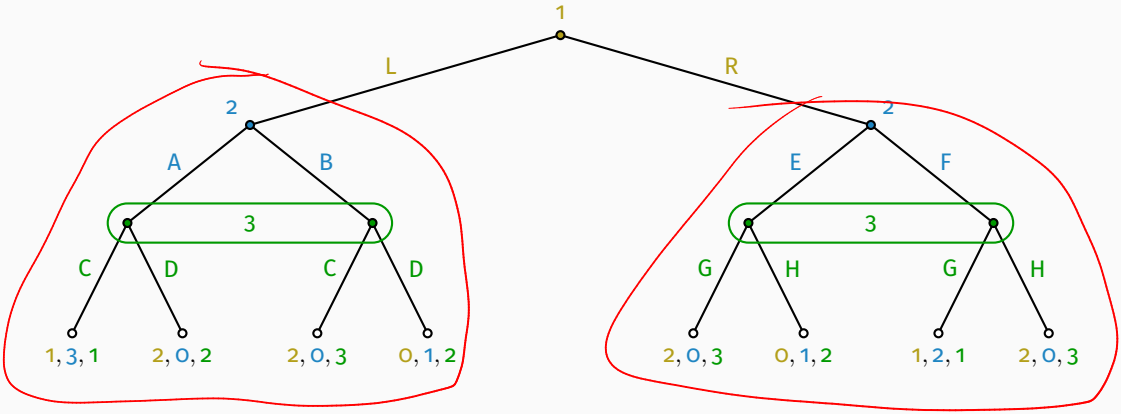
- **Security**: LAX's ARMOR software draws each day's checkpoint schedule
- **Inspections**: random drug-test lists, tax audits, restaurant inspections
- **Exams**: a random version of the test, handed out before it starts

Kuhn: with perfect recall, the two are equally powerful

A Bigger Example

Figure 7.7

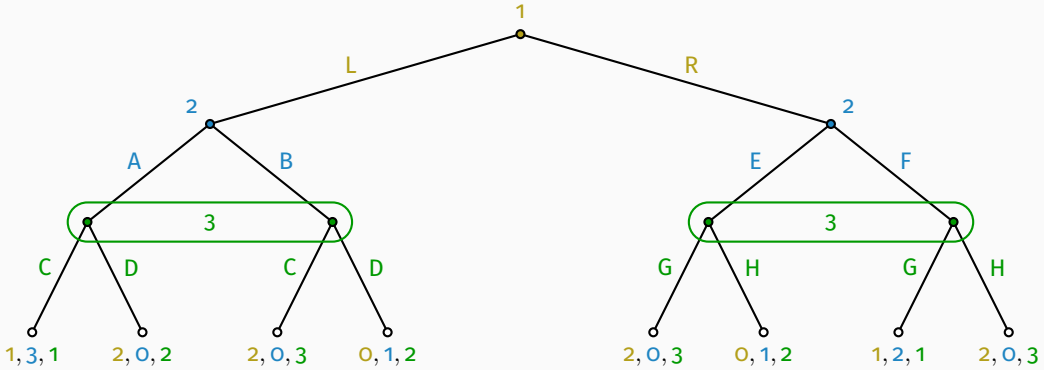
Find the SPNE



How many proper subgames? _____ Any pure NE in them?

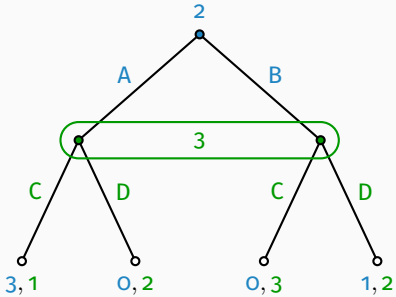
Figure 7.7

Find the SPNE



How many proper subgames? 2 Any pure NE in them? No!

Figure 7.8: The Left Subgame



Player 3

Player 2	C	D
A	3, 1	0, 2
B	0, 3	1, 2

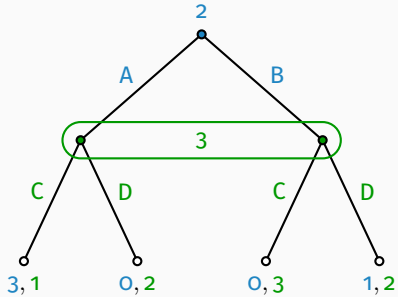
$q = \Pr(C)$ makes P2 indifferent:

$p = \Pr(A)$ makes P3 indifferent:

Payoffs (P1, P2, P3): (_____, _____, _____)

P1's payoffs come from the tree

Figure 7.8: The Left Subgame



Player 3

Player 2	C	D
A	3, 1	0, 2
B	0, 3	1, 2

$q = \Pr(C)$ makes P2 indifferent:

$3q = 1 - q \;\Rightarrow\; q = \frac{1}{4}$

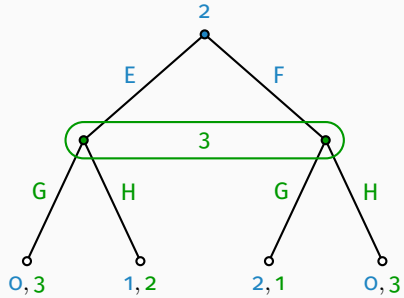
$p = \Pr(A)$ makes P3 indifferent:

$p + 3(1-p) = 2 \;\Rightarrow\; p = \frac{1}{2}$

Payoffs (P1, P2, P3): $\left(\ \frac{9}{8}\ ,\ \frac{3}{4}\ ,\ 2\ \right)$

P1's payoffs come from the tree

Figure 7.10: The Right Subgame



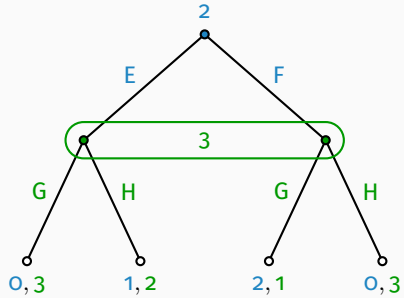
Player 2 \ Player 3	G	H
E	0, 3	1, 2
F	2, 1	0, 3

$q = \Pr(G)$ makes P2 indifferent:

$p = \Pr(E)$ makes P3 indifferent:

Payoffs (P1, P2, P3): (_____, _____, _____)

Figure 7.10: The Right Subgame



Player 2 \ Player 3	G	H
E	0, 3	1, 2
F	2, 1	0, 3

$q = \Pr(G)$ makes P2 indifferent:

$1 - q = 2q \;\Rightarrow\; q = \frac{1}{3}$

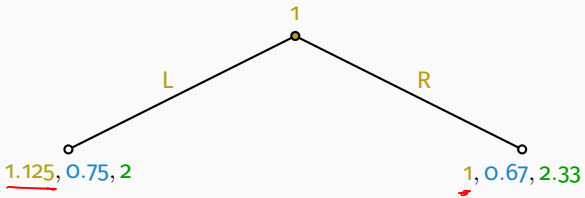
$p = \Pr(E)$ makes P3 indifferent:

$3p + (1-p) = 2p + 3(1-p) \;\Rightarrow\; p = \frac{2}{3}$

Payoffs (P1, P2, P3): $\left(\ 1\ ,\ \frac{2}{3}\ ,\ \frac{7}{3}\ \right)$

Figure 7.11: What's Left

SPNE



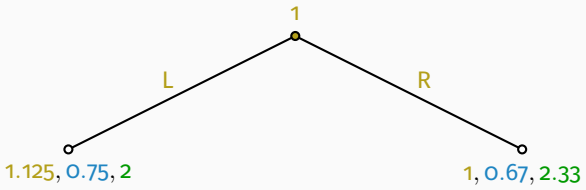
Player 1 chooses: L

SPNE, in behavioral strategies:

P1: $\begin{pmatrix} L & R \\ __ & __ \end{pmatrix}$ P2: $\left(\begin{array}{cc|cc} A & B & E & F \\ __ & __ & __ & __ \end{array}\right)$

P3: $\left(\begin{array}{cc|cc} C & D & G & H \\ __ & __ & __ & __ \end{array}\right)$

Figure 7.11: What's Left



Player 1 chooses: L

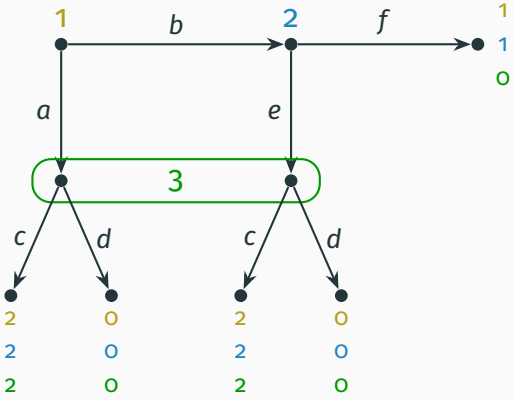
SPNE, in behavioral strategies:

$$\text{P1:} \begin{pmatrix} L & R \\ 1 & 0 \end{pmatrix} \qquad \text{P2:} \left(\begin{array}{cc|cc} A & B & E & F \\ \frac{1}{2} & \frac{1}{2} & \frac{2}{3} & \frac{1}{3} \end{array}\right)$$

$$\text{P3:} \left(\begin{array}{cc|cc} C & D & G & H \\ \frac{1}{4} & \frac{3}{4} & \frac{1}{3} & \frac{2}{3} \end{array}\right)$$

A Critique of SPNE
Due to Selten Himself

A Critique of SPNE: Selten's Horse

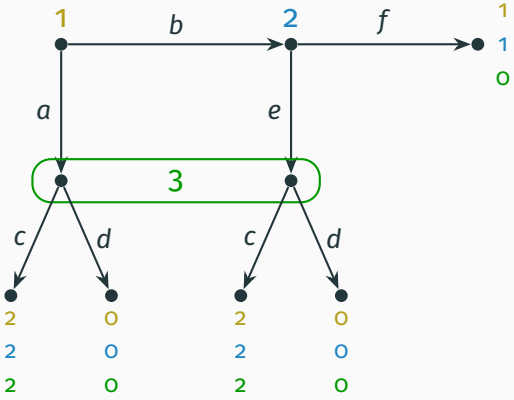


only subgame is the game itself

Proper subgames? ○

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

A Critique of SPNE: Selten's Horse

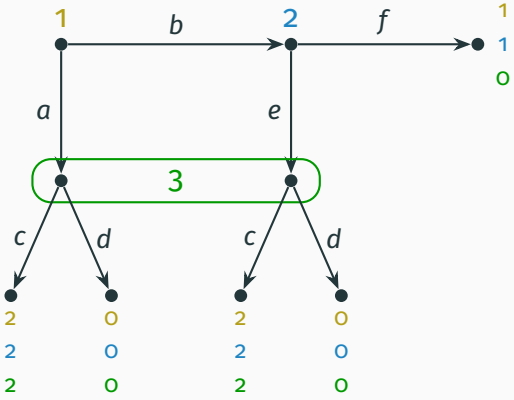


Proper subgames? _____

So SPNE = NE

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

A Critique of SPNE: Selten's Horse



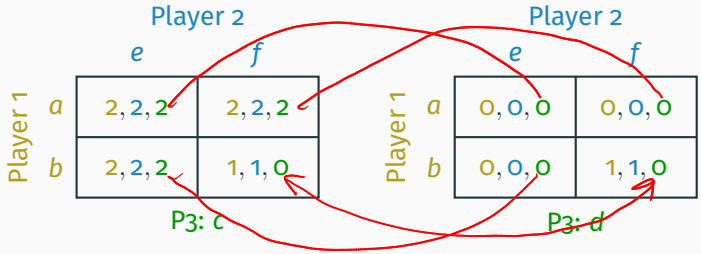
Proper subgames? _____

So SPNE = NE

To find the NE: write the strategic form

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

Selten's Horse: Strategic Form



P1 picks the row, P2 the column, P3 the matrix

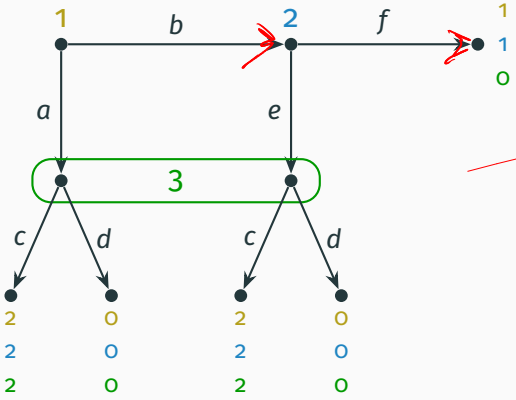
Selten's Horse: Strategic Form





Pure NE: (a,e,d) (a,f,c) (b,e,c) (b,f,d) = SPNE

Selten's Horse



Pure NE (all are SPNE):

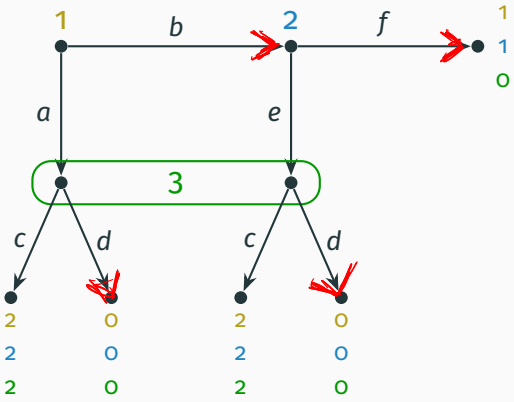
(a, e, c) (a, f, c)

(b, e, c) (b, f, d)

Are they all reasonable???

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

Problem: the (b, f, d) SPNE

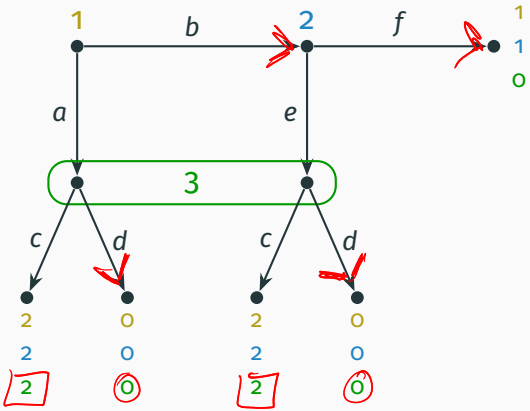


Consider the (b, f, d) SPNE

P1 plays b, P2 plays f. P3 never moves

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem: the (b, f, d) SPNE



Consider the (b, f, d) SPNE

P1 plays b, P2 plays f. P3 never moves

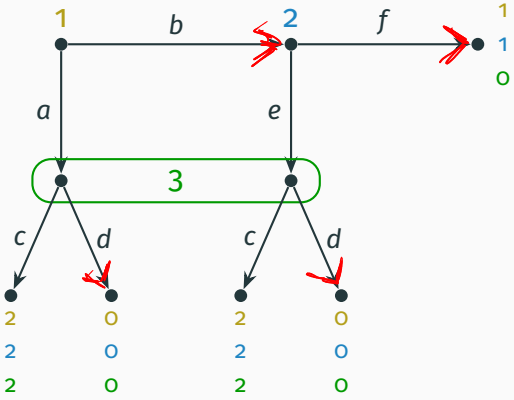
But **if** P3 had to move?

Left node: c 2 vs. d 0

Right node: c 2 vs. d 0

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem: the (b, f, d) SPNE



Consider the (b, f, d) SPNE

P1 plays b, P2 plays f. P3 never moves

But **if** P3 had to move?

Left node: c _____ vs. d _____

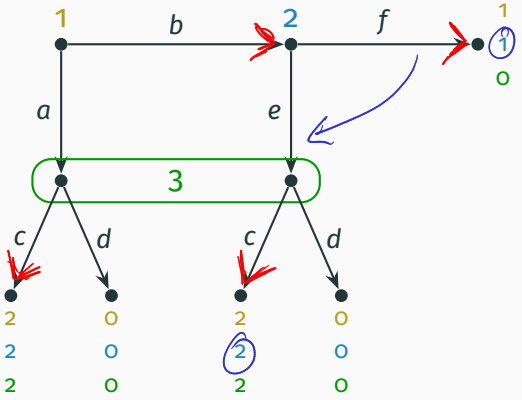
Right node: c _____ vs. d _____

d is **strictly dominated** at P3's infoset!

d is a non-credible threat
SPNE is supposed to
remove those!

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

Problem: the (b, f, d) SPNE



Consider the (b, f, d) SPNE

P1 plays b, P2 plays f. P3 never moves

But **if** P3 had to move?

Left node: c _____ vs. d _____

Right node: c _____ vs. d _____

d is **strictly dominated** at P3's infoset!

If 2 knows 3 will play d, 2 switches to e

Problem: Player 3's info set isn't the root of a subgame. In this SPNE it's never reached, so anything is a best reply

Bonanno typo, p. 249: for (b, f, d) it says "Player 1 plays a". Should be b.

So What's Missing?

SPNE checks for rationality only in **subgames**

- A subgame can't cut an information set
- Maybe that's not strict enough?

The fix: Check that strategies are rational at **every** infoset (not just subgames)

Goal: $\{\text{better solution concept}\} \subseteq SPNE \subseteq NE$

Coming in Chapter 11: a new solution concept that fixes this

- **Weak sequential equilibrium**:
  1. Players have beliefs over their infosets (*e.g.*, Pr(right node)=1)
  2. Require players best respond to their beliefs at every infoset (*e.g.*, play *c*)

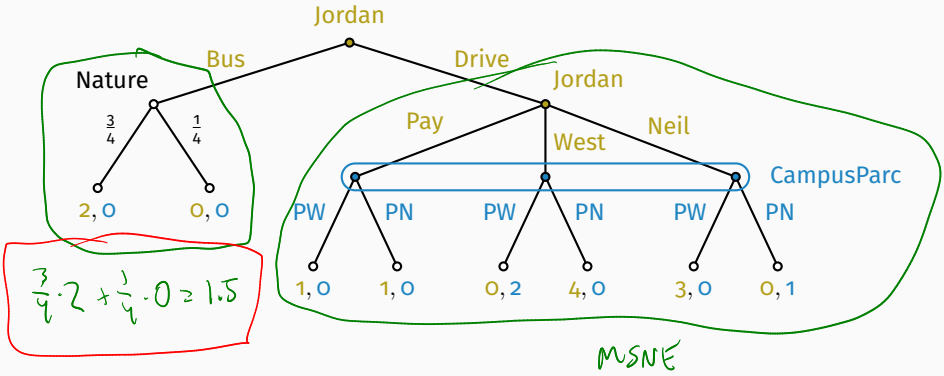
Another fix: **trembling-hand perfect equilibrium** (Selten, 1975). Not covered in this class.

Let’s Work One Out

Gameday Parking, Part 2

Jordan first decides how to get to the game:

- **Bus** (COTA): on time with chance $\frac{3}{4}$ (payoff 2), else late (payoff 0)
- **Drive**: then Chapter 6's parking game. CampusParc can't see where Jordan parked



Gameday Parking: The Drive Subgame

Jordan \ CampusParc	Patrol West	Patrol Neil
Pay	1, 0	1, 0
West	0, 2	4, 0
Neil	3, 0	0, 1

The subgame after Drive. Same as Chapter 6's parking game

Gameday Parking: The Drive Subgame

Jordan \ CampusParc	Patrol West	Patrol Neil
Pay	1, 0	1, 0
West	0, 2	4, 0
Neil	3, 0	0, 1

No cell is fully underlined: no pure NE. Need the mixed NE (Chapter 6)

Gameday Parking, Part 2: Answers

Jordan's pure strategies: _____ Numbers for a mixed strategy: _____ Behavioral: _____

Expected payoff from Bus: _____

Subgame NE: Jordan $\begin{pmatrix} \text{Pay} & \text{West} & \text{Neil} \\ ____ & ____ & ____ \end{pmatrix}$ CampusParc $\begin{pmatrix} \text{PW} & \text{PN} \\ ____ & ____ \end{pmatrix}$

Subgame payoffs: (_____, _____) Jordan chooses: _____

SPNE: Jordan $\left(\begin{array}{cc|ccc} \text{Bus} & \text{Drive} & \text{Pay} & \text{West} & \text{Neil} \\ ____ & ____ & ____ & ____ & ____ \end{array}\right)$ CampusParc $\begin{pmatrix} \text{PW} & \text{PN} \\ ____ & ____ \end{pmatrix}$

Bus is worth it if it's on time with chance at least _____

Gameday Parking, Part 2: Answers

Jordan's pure strategies: 6 Numbers for a mixed strategy: 5 Behavioral: 3

Expected payoff from Bus: $\frac{3}{2}$

Subgame NE: Jordan

Pay	West	Neil
0	$\frac{1}{3}$	$\frac{2}{3}$

CampusParc

PW	PN
$\frac{4}{7}$	$\frac{3}{7}$

Subgame payoffs: ($\frac{12}{7}$, $\frac{2}{3}$) Jordan chooses: Drive

SPNE: Jordan

Bus	Drive	Pay	West	Neil
0	1	0	$\frac{1}{3}$	$\frac{2}{3}$

CampusParc

PW	PN
$\frac{4}{7}$	$\frac{3}{7}$

Bus is worth it if it's on time with chance at least $\frac{6}{7}$

What Have We Learned, Charlie Brown?

- Nature in a tree? Replace the lottery with its **expected utility**
- **Behavioral strategy**: one probability distribution per information set
  - Mixed: spin one wheel over whole plans. Behavioral: flip as you go
- **Kuhn (1953)**: with perfect recall, behavioral $=$ mixed
- **Selten (1965)**: with cardinal payoffs, a SPNE always exists
  - Solve subgames with mixed NE when there's no pure one
- SPNE can still allow silly choices at unreached infosets (Selten's horse)
- The fix comes in Chapter 11: weak sequential equilibrium

Reading & Practice

- Bonanno, *Game Theory*, Chapter 7
  - §7.1: read for behavioral strategies and Kuhn's theorem. Skim the formal "game with cardinal payoffs" part
  - §7.2–7.3: read carefully
- Exercises: §7.4.1 (7.1, 7.2, 7.3), §7.4.2 (7.5, 7.7)
  - 7.4 and 7.6 review Chapter 5's utility calibration
  - 7.8 is harder. 7.9 (the mugger) is a challenge
- Solutions in §7.5. Try them first